



Personalized e-Commerce

Amazon.com case study

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Outline

- ☛ Technology trends
- ☛ Amazon.com example
- ☛ Research areas
 - rethinking marketing
 - rethinking business metrics & their measurement
 - rethinking user behavior studies
 - rethinking privacy
- ☛ Concluding remarks

Technology Trends

- ☞ Internet growth
 - Faster than any other infrastructure
- ☞ Data collection
 - Rapid drop in storage costs
 - Dramatic improvement in resolution and rate of data collection 'probes'
- ☞ Data analytics
 - Increasing deployment of warehouses
 - Major leap forward in data mining technologies and tools

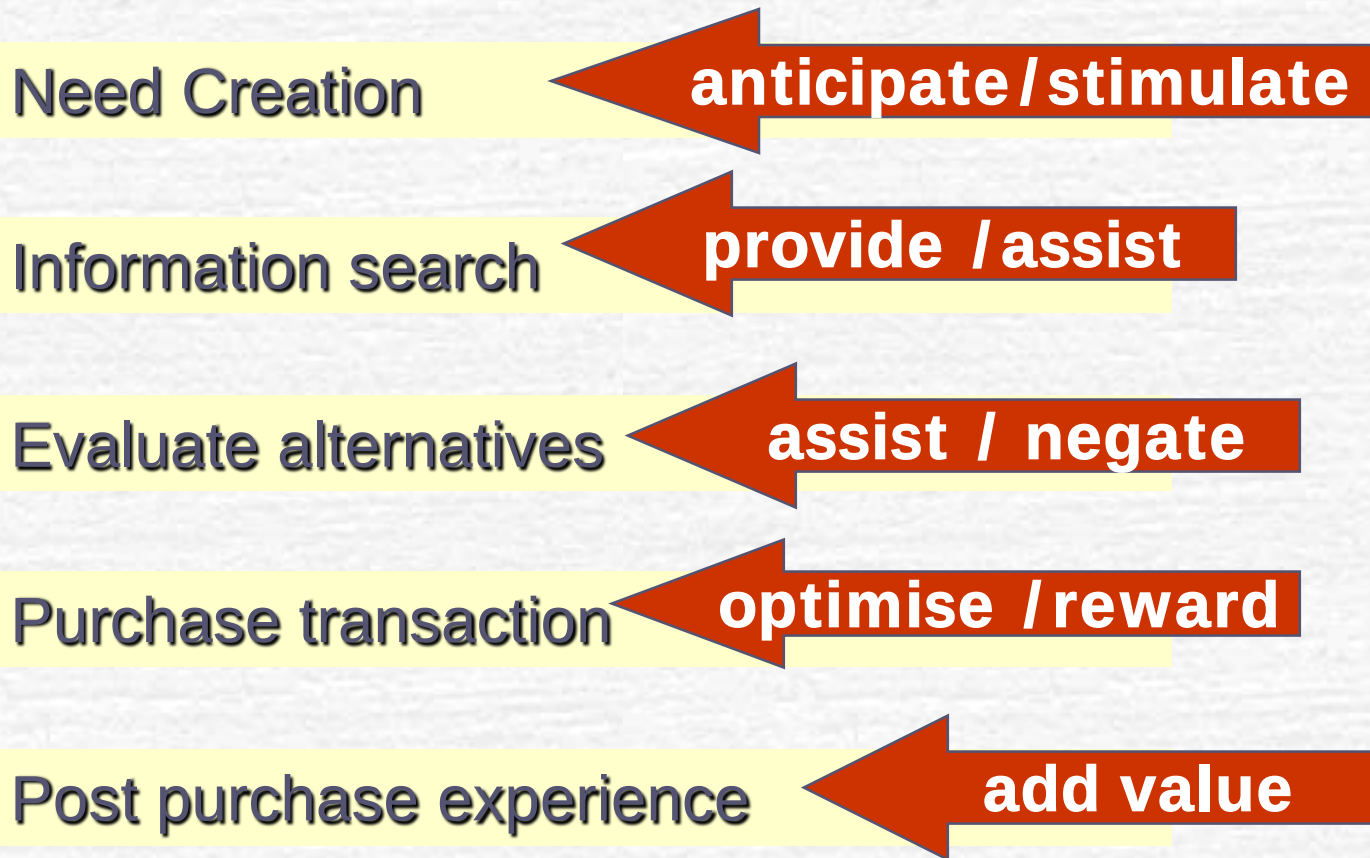
Becoming possible to really understand what is going on in the business and in interaction with customers



Amazon.com Example

The continuing relationship ...

Amazon.com “Loyalty” model



Need Creation (attract to website)

Need Creation

anticipate/stimulate

From: Amazon.com [music-store-news@amazon.com]
To: dick@dickstr.demon.co.uk
Cc:
Subject: Anne-Sophie Mutter's "The Four Seasons"

Dear Amazon.com Customer,

We've noticed that many of our customers who've purchased CDs by Yo-Yo Ma also enjoy music by violinist Anne-Sophie Mutter. For this reason, you may like to know that Mutter's new recording of Vivaldi's "The Four Seasons" has just hit the shelves. For the next few days, you can order your copy at a savings of 30% by following the link below:

<http://www.amazon.com/exec/obidos/ASIN/B00002DE2L>

Further Need Creation (upon reaching website)

Browse Purchase Circles

- [Geography](#)
Countries, States, Cities...
- [Government](#)
Independent, Military...
- [Organizations](#)
Nonprofit, Professional...
- [Companies A-Z](#)
Business, Entertainment...
- [Education A-Z](#)
Colleges, Universities...

Information Search

Information search

provide / assist



SEARCH

All Products ▼

GO!

Search of the Day:
[Buena Vista Social Club](#)

All Products

Popular Music

Classical Music

DVD & Video

Toys & Games

Electronics

Software

Home Improv.

Auctions

zShops

sothebys.amazon

Evaluation of Alternatives

Evaluate alternatives

assist / negate

The image shows a screenshot of a Barnes & Noble (bn.com) website. A large, rounded rectangular pop-up window is overlaid on the page. The pop-up has a 'Beta' badge and the 'zBubbles amazon.com' logo. It contains the text 'Be the first to tell other shoppers about a product on this page...' followed by a bulleted list of suggestions: 'another place to buy it', 'a related product', and 'buying it at Amazon.com'. Below the list is the text 'or start shopping here...'. At the bottom of the pop-up is a search bar with a dropdown menu set to 'All Products' and a 'GO!' button. The Amazon.com logo and tagline 'The place to find & discover anything you might want to buy online!' are also present, along with a 'Close' button.

bn.com
BARNES & NOBLE

Books Music eCards Prints & Posters Software Magazine

Browse Kids

January

RE
YOU

A new m
a new yo

BUY \$
Need He
Mo
Dec

Special
Oprah's
Gap Cree
bn.com L
Award Winn
Bestsellers

Beta
zBubbles
amazon.com

SEARCH | HELP
YOUR ACCOUNT SETTINGS

Be the first to tell other shoppers
about a product on this page...

- [another place to buy it](#)
- [a related product](#)
- [buying it at Amazon.com](#)

or start shopping here...

Search: All Products GO!

amazon.com The place to find & discover
anything you might want
to buy online!

Close

Purchase Optimisation/Reward

Purchase transaction

optimise / reward

- 1-click purchase
- 'slippery check out counter' vs. 'sticky aisles'



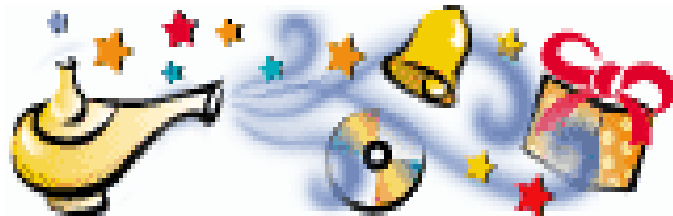
WELCOME TO
Amazon.com Refer-A-Friend

Invite Your Friends to Amazon.com, and We'll Thank You Both

Post-purchase experience

Post purchase experience

add value



Wish List



Your 1-Click and
Gift-Click Settings

Rethinking Marketing

☞ What has changed

- Ability to engage customers in a near real-time, interactive, personalized dialog
- Comprehensive customer profiles
- Detailed recording of customer behavior

☞ What can this lead to

- Multi-round, interactive price negotiation
- Customer segmentation
- Developing detailed understanding of customer preferences for
 - Sales
 - Product development

Marketing – 75 years ago

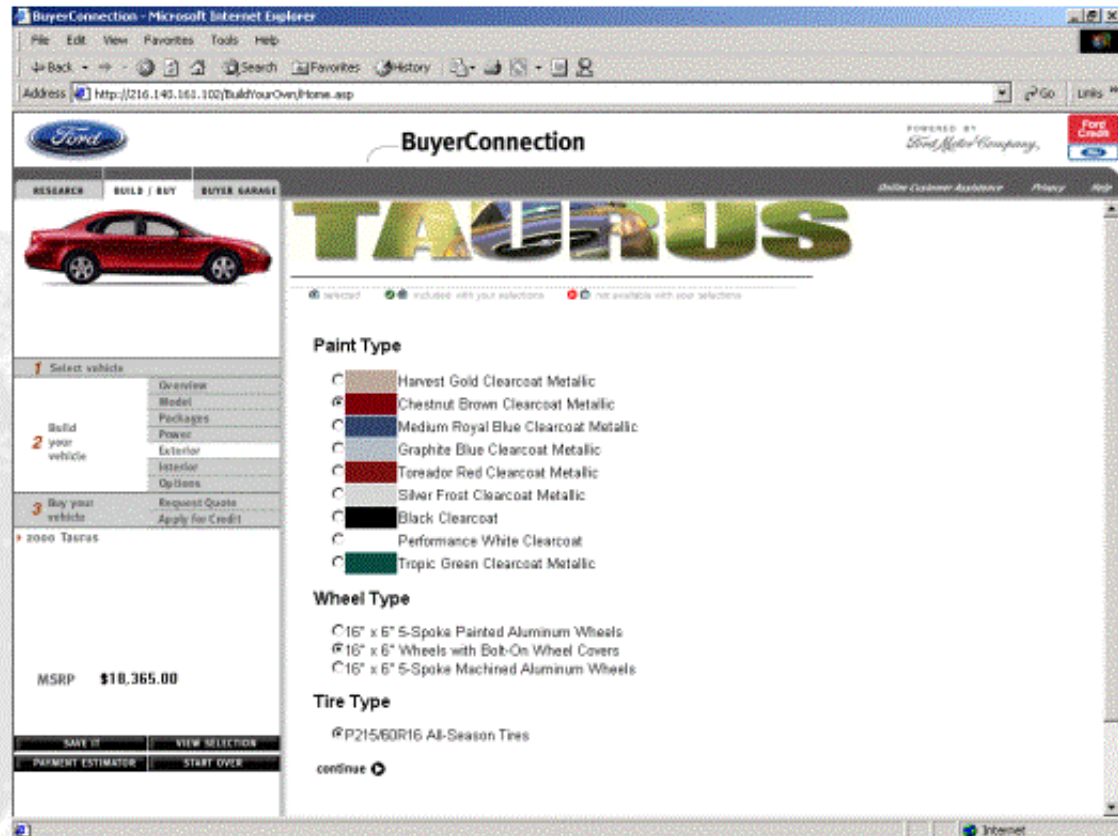
- Production – a la Adam Smith
- You can have any color as long as it's black – Ford Motor Co.



CRM-Forum 4

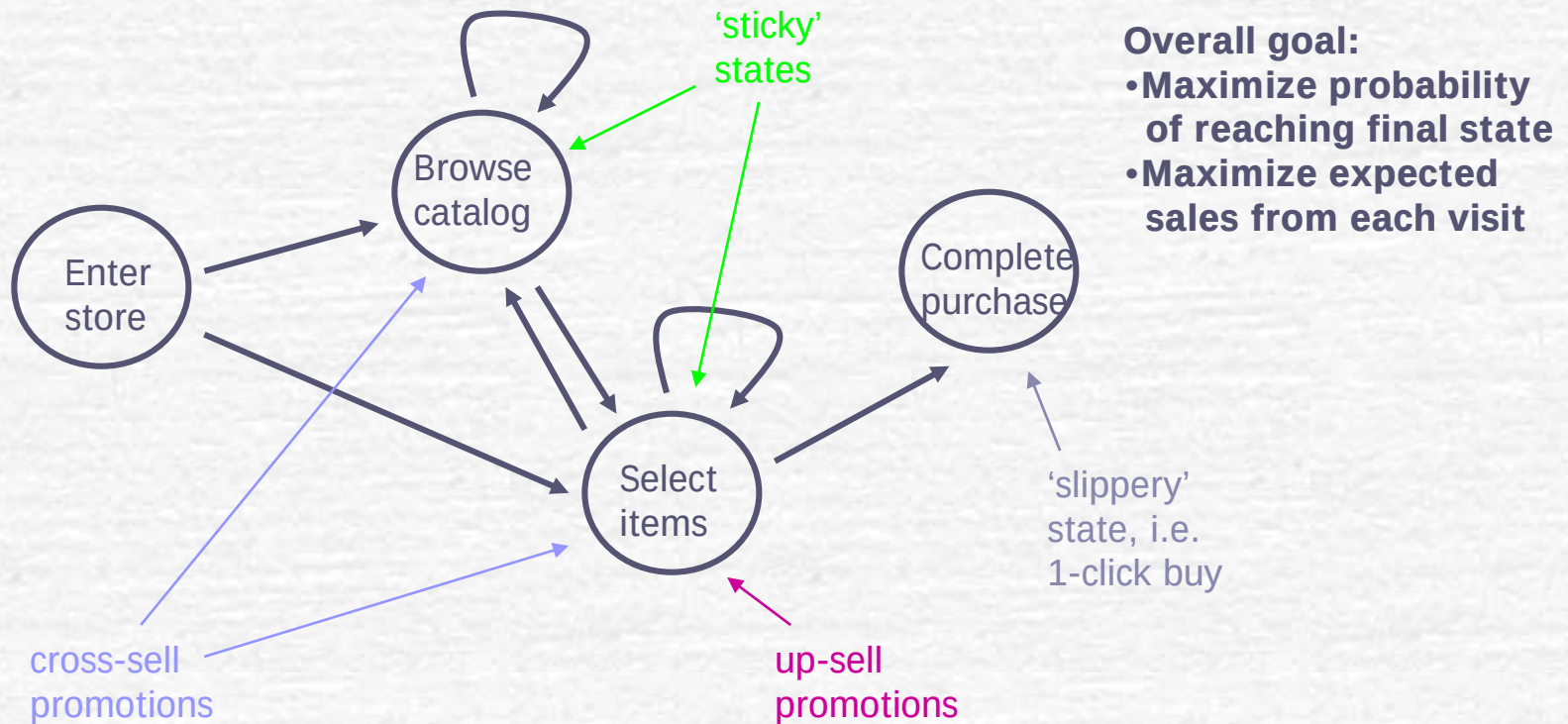
Marketing - today

Add the spice
of flexibility,
courtesy of
robotics,
computers ...



CRM-Forum

Rethinking Business Metrics



- Shopping pipeline modeled as state transition diagram
- Sensitivity analysis of state transition probabilities
- Promotion opportunities identified
- E-metrics and ROI used to measure effectiveness

Rethinking User Behavior Studies

☞ Amazon.com's WebLab

- The web a lab
- Customers as subjects
- Advantage over standard user studies
 - Much much larger scale – millions
 - Users don't know they are participating in a study

☞ New issues

- Design of web lab experiments
- Measurement and analysis of behavior

Rethinking Privacy

- ☛ Is it 'much ado about nothing'?
 - If indeed data collection was outlawed, and thus personalization impossible, wouldn't the public lose – faced with generic, undifferentiated products/services?
 - Given the public's attitude about privacy (as shown in their actions), are privacy advocates barking up the wrong tree?
 - Is it just a matter of time or generational issue, e.g. adoption of credit cards
- ☛ Where do we stand?
 - Current position - loss of your privacy may be beneficial for you
 - Emerging position (post September 11th) - loss of your privacy will be beneficial for everyone
 - Critical emerging debate - is **privacy a right or a privilege?**

Concluding Remarks

- Technology trends are creating an unprecedented opportunity to study business processes and customer behavior
- Data collection and data analysis technologies are advancing at a rapid rate
- New research areas have opened up, and researchers in business and engineering schools must work together

let's talk about privacy ...

☞ Merriam Webster definition

- **a:** the quality or state of being apart from company or observation **b :** freedom from unauthorized intrusion

☞ Justice Oliver Wendell Holmes

- “the right to be left alone”

☞ Operational definition

- Collection and analysis of personal data beyond some limit

Public Attitude Towards Privacy

- ☛ A (self-professed) non scientific study carried out by a USA Today reporter
- ☛ Asked 10 people the following two questions
 - Are you concerned about privacy? 8 said YES
 - If I buy you a Big Mac, can I keep the wrapper (to get fingerprints)? 8 said YES
- ☛ ACM E-Commerce 2001 paper [Spiekermann et al]
- ☛ Most people willing to answer fairly personal questions to anthropomorphic web-bot, even though not relevant to the task at hand
- ☛ Different privacy policies had no impact on behavior
- ☛ Study carried out in Europe, where privacy consciousness is (presumably) higher

Public Attitude (contd.)

- ☛ Amazon.com (and practically every commercial site) uses cookies to identify and track visitors
 - 97.6% of Amazon.com customers accepted cookies
- ☛ Airline frequent flier programs with cross promotions
 - We willingly agree to be tracked
 - Get upset if the tracking fails!
- ☛ Over 1.5 million people have trusted the aggregation service (called Yodlee) with the names and passwords of their financial accounts in less than 18 months
- ☛ Adoption rate has been over 3 times the most optimistic projections

Medical data is (perhaps) an exception to this

What people really want

- ☛ Some people will not share any kind of private data at any cost – the ‘paranoids’
- ☛ Some people will share any data for returns – the ‘Jerry Springerites’
- ☛ The vast majority in the middle wants
 - a reasonable level of comfort that private data about them will NOT be misused
 - Tangible and compelling benefits in return for sharing their private data – Big Mac example, frequent flier programs

Mantra to Summarize the Public Attitude*

☞ You can get

- none of the data some of the time (the “paranoids”)
- and all of the data some of the time (the “Springerites”)

☞ But, you can get

- most of the data most of the time

* With apologies to President Lincoln



That's all folks!!

Questions?
Comments?
Rotten eggs?